

Balancing transparency and control: The impact of AI explanation detail on user perception in automated vehicles

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ABSTRACT

As automated vehicle technology advances, explainable AI has emerged as a critical tool to enable users to understand and predict the behavior of AI systems, particularly in safety-critical applications such as automated driving. However, increased transparency in AI explanations may inadvertently contribute to an “illusion of control”, a cognitive bias in which drivers overestimate their influence or understanding of the AI’s actions. We aim to better understand how the level of detail in AI explanations affects users of automated vehicles. In a virtual reality study, $N = 44$ participants experienced different explanation levels (low, medium, high) in an automated ride (SAE L4) compared to a baseline condition with no explanations. The results show a significant improvement in participants’ user experience, acceptance, and explanation satisfaction, with more detailed explanations. Our findings also indicate that as AI explanations become more detailed, users’ perceived level of control increases significantly, although this perception does not correlate with actual control capabilities. At the same time, it decreased their desire to take control, indicating users’ susceptibility to the ‘illusion of control’ bias in the context of automated driving. Overall, this suggests that the design of explanation interfaces should strive for a balanced level of detail that promotes AI transparency without causing cognitive overload. At the same time, explainable AI can be utilized to decrease users’ desire to intervene in the AI’s actions.

1. Introduction

While automated mobility promises enhanced safety, increased efficiency in traffic flow, and more convenience for passengers, the transition from manual to fully Automated Vehicles (AVs) introduces challenges in the research domain of human factors, particularly such as overreliance (Dunn et al., 2021), skill degradation (Endsley and Kiris, 1995), and trust miscalibration (Lee and See, 2004a) because humans struggle to maintain attention, interpret automation behavior, and adapt to shared control with machines. These challenges stem from psychological biases, poor interface design, and mismatches in expectations between humans and AVs. One critical factor that influences passengers’ willingness to relinquish control to an AV is how well the automated system explains its decisions and actions to the humans on board, also referred to as Artificial Intelligence (AI) explanations (Gunning and Aha, 2019).

In the context of AVs, AI explanations encompass real-time communication of driving decisions. This includes the reasoning behind route choices, responses to dynamic traffic conditions, as well as general reactions to unexpected traffic events. Particularly focusing on conveying an AV’s decision and behavior in an explainable, transparent, and

comprehensive way to passengers through Human–Machine Interfaces (HMIs) contributes positively to trust building and acceptance (Stepin et al., 2021). Related research found that designing appropriate AI explanations is essential for fostering users’ understanding of the automation’s behavior (Koo et al., 2014). While too little information leads to a lack of trust (Manger et al., 2023a; Omeiza et al., 2021). Too much, or the wrong kind of information, on the other hand, can cause cognitive overload (Koo et al., 2014).

Another critical factor that can influence users’ trust in AVs is the perceived level of control they experience in driving situations. Increasing levels of automation are inherently associated with a loss of control, and contrary to expectations, recent studies demonstrate that passengers in highly and fully automated vehicles still report a continued desire for control over driving tasks (Malve et al., 2023a; Peintner et al., 2023a). However, it is unclear whether this desire for control stems from an inherent lack of trust and acceptance towards the AV due to inadequate AI explanations or whether passengers inside AVs inherently have an intrinsic desire to alter the automation’s behavior and decisions to suit their preferences.

Hence, this research aims to explore the intricate relationship between the explainability of AV systems and passengers’ desire to control

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the vehicle. At the same time, we aim to better understand *which* and *how much* information passengers and drivers in AVs require. By utilizing an explanation HMI, we research how these factors influence passengers' desire to control the vehicle. Hence, we focus on the type and detail of explanations given by the automation while also investigating their influence on users' perception of the vehicle.

To this end, we conducted a virtual reality driving simulation experiment in which $N = 44$ participants experienced different HMI-based AI explanation strategies while encountering various traffic situations. Our insights demonstrate that the level of explanation has a significant impact on passengers' control perception, as well as their perceptions of the automation in terms of *User Experience*, *Perceived Safety*, *Trust*, and *Acceptance*. These combined insights contribute to a deeper understanding of human–AI interaction and offer guidance for the design of future explanation interfaces in automated vehicles.

2. Related work

In the following, we provide an overview of the current state-of-the-art research in the field of human factors in Automated Driving (AD), with a detailed view of challenges, including *User Experience*, *Perceived Safety*, *Trust*, *Acceptance*, as well as users' desire for control and perceived level of being in control. We conclude with a discussion of whether, and to what extent, AI explanations can help address these challenges.

2.1. Human factor aspects in automated driving

The users' *Acceptance* plays a pivotal role in the adoption of AVs, as it encompasses the willingness of users to embrace and integrate AD technology into their daily lives (Kyriakidis et al., 2015). *Acceptance*, as a broader concept, is influenced by various factors, such as *Trust in Automation* (Lee and See, 2004a), *User Experience* (Frison and Riener, 2022), *Perceived Safety* (Prasetyo and Nurliyana, 2023), and the perceived benefits over conventional vehicles. A user's *Trust in Automation* influences the individual's willingness to adapt to AVs. It can be defined as “the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability.” (Lee and See, 2004a, p.54). The process of building trust was summarized by Wintersberger et al.: “Trust is built on the possibility to observe the system's behavior (performance), understand the intended use of the system (purpose), as well as understand how it makes its decisions (process)” (Wintersberger et al., 2016, p.2). Users often fail to rely appropriately on automation, either overestimating its capabilities (overtrust) or underestimating them (distrust). Overtrust in AVs, in particular, has resulted in multiple accidents, such as users over-relying on Tesla's Autopilot system (Hawkins, 2024). Muir and Moray introduced the terms “misuse” and “disuse” to describe the improper reliance on automation due to overtrust and distrust, respectively (Muir and Moray, 1996). To address these issues, research has emphasized the importance of *Calibrated Trust*, which refers to the alignment between a user's level of trust and the system's actual capabilities, limitations, and performance (Lee and See, 2004b). This alignment helps prevent both overtrust (misuse) and undertrust (disuse), enabling users to rely on the system when it is competent while remaining vigilant in situations where it may fail. Therefore, adequately calibrated trust serves as a foundation for safe human-automation interaction, ensuring that reliance on AVs is neither excessive nor insufficient. Empirical studies investigated how calibrated trust emerges and found a strong influence of clearly communicated system capabilities and experience of the system. For example, uncertainty displays (Helldin et al., 2013; Kunze et al., 2019), or transparent, video-based feedback (Wintersberger et al., 2021) lead users to calibrate their trust levels in line with the system reliability. Furthermore, simulator and on-road exposure (Walker, 2021; Clement et al., 2022) was shown to help users reduce initial overtrust by allowing repeated observation of system limitations and variability.

User Experience is another critical factor in AD as it directly impacts overall acceptance and satisfaction with the technology or system used (Riener, 2018). Factors influencing a product's or system's *User Experience* include attractiveness, perspicuity, efficiency, dependability, stimulation, and novelty (Schrepp et al., 2017). By prioritizing *User Experience*, researchers and designers can create AD systems that not only meet functional requirements but also address the emotional and psychological needs of users, as described in the following.

(*Perceived*) *Safety* in AD is strongly related to users' *Trust* and *Acceptance* of the technology. It refers to the users' subjective assessment of how safe they feel while interacting with or being transported by an AV, which encompasses the users' *cognitive* and *emotional* safety (Prasetyo and Nurliyana, 2023). It also includes their confidence in the vehicle's ability to handle various driving situations, avoid accidents, and ensure the well-being of its occupants. When users feel safe, they are more likely to rely on and effectively utilize AD systems (Zoellick et al., 2019).

2.2. Cognitive biases: Sense of control and the illusion of control

Sense of Control refers to the perceived level of influence or power an individual has over a situation or a personal event (Bandura, 1977). It is considered an important quality of human–machine interaction with a strong impact on *User Experience* and *Usability* and, therefore, included in classic interface design principles, for example, by Shneiderman et al. (2016) and Nielsen (1994). Control beliefs are named differently in the various psychological models, e.g. *Self-Efficacy* (Bandura, 1977), or *Perceived Behavioral Control* (Ajzen, 1985), but describe overall a similar concept.

In the context of AD, surrendering control to automation remains one of the main human factor challenges in the adoption of AVs, resulting in various approaches to counter the loss of control, such as tailoring the AD style to the user (Hartwich et al., 2018). Syahrivar et al. (2021) found that a more positive attitude toward AVs is associated with a stronger intention to use them. At the same time, a higher desire for control correlates with a more negative attitude toward AVs, and consequently, a lower intention to adopt them (Syahrivar et al., 2021). Weigl et al. (2022) conducted questionnaire studies on acceptance of AVs and the desire for control. A factor analysis found only a weak association between the desire for control and acceptance of AVs, and the authors encourage further investigation.

Research has found that a binary control-allocation approach following the SAE definition (SAE On-Road Automated Driving (ORAD) Committee, 2021) to designing interactions with AVs is not ideal from a human factors perspective (Guo et al., 2019; Walch et al., 2017; Wiegand et al., 2020a). Next to the known issues with human operators taking back control after being out of the control loop for extended periods (Endsley and Kiris, 1995), research has also shown that passengers in highly and fully automated vehicles still tend to want control over the vehicle, even though this is not required, solely based on the automation's capabilities (Wolf, 2016; Malve et al., 2023b; Peintner et al., 2023a).

Hence, the feeling of being in control is a crucial aspect of human psychology that is even associated with a *Cognitive Bias* known as the *Illusion of Control*. *Cognitive Biases* are systematic errors in rational judgment and thinking that affect human perception and decision-making. These biases can arise from mental shortcuts or heuristics that our brains employ to expedite decision-making, particularly under conditions of uncertainty or information overload (Tversky and Kahneman, 1974).

Illusion of Control is a *Cognitive Bias* that causes individuals to overestimate their ability to influence events or outcomes that are primarily, if not entirely, determined by chance or external factors. The concept was introduced and demonstrated through a series of experiments by psychologist Ellen Langer in the 1970s (Langer, 1975). *Illusion of Control* often occurs in gambling (Eben, 2025), financial

decisions (Nicholson et al., 2003), or sports (Burger and Cooper, 1979). The bias is more substantial when elements of skill, choice, or familiarity are present (Langer, 1975). The bias-induced subjective belief in personal control over a situation often results in an inflated self-assessment of one's actual influence on a situation, which can lead to overconfidence and increased risk-taking (Yarritu et al., 2014).

In the context of AD, early evidence suggests that *the Illusion of Control* can occur situationally. A study by Koo et al. (2016) found that providing drivers with additional information via voice alerts during a fully automated ride increased their *Sense of Control*, even though their actual level of control over the situation remained unchanged compared to a condition without such alerts. Similarly, Manger et al. (2023b) observed in a user study that a greater *Sense of Control* emerged when participants were given information about the behavior of a fully automated vehicle. Schneider et al. (2023) found that transparency increases *Sense of Control* in users of AVs, which in turn improves their user experience. The authors introduce the concept of '*Information Dilemma*': while increased transparency generally improves user experience under normal conditions, it can backfire in failure scenarios by exposing system limitations without offering users any means of intervention, potentially degrading the overall user experience.

2.3. AI explanations in automated vehicles

Explainability in AD systems addresses the described human factor challenges by calibrating users' *Trust*, comprehension, and interaction with these technologies. AI systems, such as Machine Learning (ML) applications, are often referred to as black boxes with opaque decision-making processes (Gunning and Aha, 2019). If users of these systems are not able to understand the decisions of such an AI system, it can negatively affect their *Trust* (Dzindolet et al., 2003; Lyons et al., 2017) and *Acceptance* of the decisions (Cramer et al., 2008; de Fine Licht and de Fine Licht, 2020).

By making the decision-making processes of AVs transparent, drivers and passengers can better understand and anticipate the system's decisions (de Fine Licht and de Fine Licht, 2020), leading to increased confidence and more effective collaboration between humans and the AV (Christoffersen and Woods, 2001; Klein et al., 2004). Omeiza et al. (2021) surveyed general research on explanations in AD and derived that explanations can be categorized based on the causal filters, the content style, the interactivity, the model dependence, the system, and the scope.

In the context of user interfaces for AD, research is ongoing regarding the suitable timing of the explanations (Schneider et al., 2021; Kim et al., 2023), the modality (Peintner et al., 2023b), the individual user's needs and preferences (Wintersberger et al., 2020), and the content and amount of information (Wiegand et al., 2020b; Danner et al., 2021). Omeiza et al. (2025) examined the effects of explanation specificity and AV perception system errors in a driving simulation experiment. They found that abstract explanations did not make passengers feel safer despite being vague enough to conceal the detection errors, compared to specific explanations. Anxiety increased with high explanation detail. Moreover, the choice of appropriate modalities for delivering explanations remains a recurring topic of discussion. Many concepts rely on visual, text-based explanations, which can effectively convey even complex information and help prevent misinterpretation. However, varying levels of automation and the associated shifts in driver visual attention may require other modalities. Peintner et al. (2023b) compared visual explanations with auditory and vibro-tactile modalities for conveying automation uncertainty. The results suggest that auditory and vibro-tactile explanations improved *User Experience*, *Trust*, and *Perceived Safety*. Notably, the visual explanation in the study was delivered via an abstract LED strip in the vehicle interior, highlighting that the form of representation plays a critical role in user perception. Kunze et al. (2018) investigated the suitability of various visual characteristics for representing automation uncertainty

in dynamic augmented reality (AR) displays. Results indicate that color-based and animation-related variables, such as hue, were most preferred by participants, suggesting their potential effectiveness in conveying uncertainty in AR interfaces. Visualization using AR can also improve *Situation Awareness* (Manger et al., 2023a,c), while also maintaining a low cognitive load (Colley et al., 2021). Stange et al. (2022) conducted two experimental studies that investigated whether passengers' trust in automation can be supported by adapting vehicle behavior to demonstrate the vehicle is "aware" of a potential hazard (e.g., pedestrian crossing the street) as an implicit explanation of its behavior. Results show that a deceleration might be an effective behavioral adaptation to inform passengers about the awareness of a hazard. Using a lateral offset as behavioral adaptation had no positive effect on passengers' perceived risk and trust ratings.

Overall, the effectiveness of these AI explanations greatly depends on the explanation content (Zhang et al., 2021), and the provided amount of information, which can be seen as the "*Level of Explainability*". Yet there is no general categorization on which explanations refer to low levels of explainability and which ones to high levels of explainability. In various application fields, an increasing amount of information describes an increasing "*Level of Explainability*", whereby too much information can overwhelm the user and does not necessarily lead to better explainability (Miller, 2019). Koo et al. (2014) first introduced the approach of separating different levels of explanation content into the categories "*How*", "*Why*", and "*How & Why*" in the context of explaining the behavior of AVs. Koo et al. (2014) and Zhang et al. (2021) found that these different levels of explanations affect the *User Experience* of the driver as well as the safety of their driving behavior. Contrary to these factual explanations, describing how and why the AI system came to a certain decision, contrastive and counterfactual explanations have been explored in the context of AI explanations. They explicitly describe why the AI system did not produce a different outcome and the conditions that would need to change for an alternative result to occur (Stepin et al., 2021). This type of explanation has the potential to improve interactions of humans with AI systems since it adds transparency and provides indications as to what actions can change the output of an algorithm (Stepin et al., 2021).

3. Research questions

To date, it is an unanswered question what the underlying reasons are for the inherent desire for control that passengers of L4 and L5 vehicles seem to have. And how a lack of *Trust* and *Acceptance* in automation affects this desire. Additionally, it is interesting to study what role users/passengers expect preferences in AV behavior to play. Current research findings regarding the "*Illusionary Control*" bias indicate that a user's sense of control can be increased by simply giving more detailed explanations about the AI's reasoning behind its decision-making (Manger et al., 2023b). This raises the question of whether passengers' desire for control can be decreased by providing more detailed explanations concerning AI decision-making.

With our research, we aim to contribute to a better understanding of the relationship between AI explanations, passengers' desire for control, and their perception of automation in terms of *Trust*, *Acceptance*, *Perceived Safety*, and *User Experience*.

Thus, we focus on answering the following research questions (RQs):

- RQ1:** How does the "*Level of Explainability*" of the automation impact the users' *Sense of Control* in terms of desire for control and perceived level of being in control?
- RQ2:** How does the "*Level of Explainability*" of the automation impact the users' perception of the vehicle in terms of *User Experience*, *Trust*, *Acceptance*, and *Perceived Safety*?

Additionally, another potentially influencing factor, the passenger's seating position, was included in the investigation. AVs that are already on the road (e.g., Waymo LLC (2025), Tesla, Inc. (2025), Cruise LLC (2025)) have a seating arrangement for passengers in line with standard taxi settings, where passengers often occupy the rear seats. In contrast, driving simulation studies usually place the participants in the driver's seat. Hence, we expect that there might be a subconscious change in passengers' driving experience when sitting in the driver's seat due to active driver role association, compared to a more passive passenger role when seated in the rear. Peintner et al. (2025) have found that the seating position has an effect on how passengers in AVs perceive traffic situations. Therefore, we formulate the following additional research questions:

RQ3: How does the *Seating Position* inside the vehicle influence participants' *Sense of Control* in terms of desire for control and perceived level of being in control?

RQ4: How does the *Seating Position* inside the vehicle influence participants' perception of vehicle automation concerning *User Experience, Trust, and Safety*?

4. Method

To investigate the four research questions outlined above, we conducted a mixed-factorial design driving simulator study with a 2×4 design in Virtual Reality (VR). To test for differences based on seating position, participants were assigned either the driver's role or the back-seat passenger's role (between-subject factor). Every participant then experienced four rides in an SAE Level 4 AV (SAE On-Road Automated Driving (ORAD) Committee, 2021). During each ride, the AV encountered four traffic situations and explained its behavior using four different levels of explanation detail: "Baseline", "How", "How & Why", and "Contrastive & Counterfactual" based on the research by Koo et al. (2014), as well as Stepin et al. (2021) (within-subject factor).

4.1. Explanation levels and HMI design

To study the underlying reasons behind users' desire for control and their perception of the vehicle without introducing design-induced biases, we opted for a visual, text-only HMI design. We placed a translucent, black text box with a written message centered and directly in the line of sight of the driver or the right-back-seat passenger. We defined four levels of explanation detail based on existing literature:

1. **No Explanation Level ("Baseline"):** No information about the vehicle's behavior was provided.
2. **Low Explanation Level ("How"):** "How" explanations were given, describing the maneuver the vehicle is currently performing, as used by Koo et al. (2014).
3. **Medium Explanation Level ("How & Why"):** "How & Why" explanations were combined (Manger et al., 2023a).
4. **High Explanation Level ("Contrastive & Counterfactual"):** A combination of "Contrastive & Counterfactual" explanations was provided (Stepin et al., 2021), detailing which actions the vehicle cannot perform and how the situation must change to maneuver successfully.

The explanations for each of the scenarios 4.2 that participants experienced in the experiment were developed based on the categories above. Three researchers developed the explanations by ideating texts for these categories similar to the related literature, which were then discussed and modified until a final text was defined. Using these explanations, the automation tries to communicate the reasoning for its current behavior. The explanation texts for each of the different explanation levels are listed in Table A.2.

4.2. Scenarios

For the virtual driving environment, we implemented a driving simulation involving an SAE Level 4 (SAE On-Road Automated Driving (ORAD) Committee, 2021) left-hand-drive AV navigating through an urban setting (Fig. 1). During this automated drive, the vehicle encountered four traffic situations requiring it to react and adapt its driving behavior.

The situations are listed in Table A.2 and are pictured in Fig. 1. The situations are based on research by Terken and Pfleging (2020), and Malve et al. (2023a), and Peintner et al. (2025), who have investigated passengers' requirements in automated mobility and derived specific use cases in which passengers might feel a need to intervene in the driving task and change the automation's behavior.

In the "Slow moving vehicle ahead" situation, the ego-vehicle encounters a slower-moving car traveling in the same lane as itself in front of it, which forces it to slow down and drive slowly along the road. As the situation is not entirely clear, the automation opts not to switch lanes, although it might be possible.

In the "Pedestrians talking and standing on the road" situation, the AV is forced to stop because of two pedestrians standing partially on the road. One of the pedestrians then tries to signal to the vehicle that it is safe to pass. However, the ego-vehicle, unable to recognize this gesture, waits until the pedestrian has stepped off the road.

Similarly, in the "Pedestrian with unclear intentions" situation, the ego-vehicle encounters a pedestrian walking towards a zebra-crossing, walking parallel to the road. Since the automation in this situation expects the pedestrian to potentially step onto the street, the vehicle stops at the zebra crossing. However, the pedestrian does not intend to cross and stops on the sidewalk close to the zebra crossing.

In the "Construction site" scenario, the ego vehicle has to stop at a construction site blocking the lane it is traveling in.

These four situations are integrated into four simulation rides, each one containing all four situations and lasting approximately five minutes. To prevent learning effects, the sequence of when the traffic situations appeared during the ride was randomized between rides, so that participants were unable to anticipate the situations.

4.3. Dependent variables

To answer our research questions, the following dependent variables were assessed using questionnaires:

User experience. *User Experience* was measured using the User Experience Questionnaire in its short version (UEQ-S) (Schrepp et al., 2017). The standardized, validated questionnaire measures efficiency, dependability, and stimulation. It consists of 8 items that capture the key aspects of *User Experience*. The ratings are based on a 7-point Likert scale, presented as semantic differentials (e.g., boring-exciting).

User acceptance. To gain an understanding of participants' *Acceptance* of the automation's explanations, we used the User Acceptance scale by van der Laan et al. (1997). With the subscales *Usefulness* and *Satisfaction*, we sought detailed insight into participants' perception of the automated system, substituting the UEQ-S scores. The scale consists of 9 items rated on a 5-point scale, e.g., ranging from "unpleasant" to "pleasant".

Trust. *Trust* in automation was measured using the single item question "I trusted the automation in this situation", which participants rated on a 5-point Likert-Scale from 1 = *not at all* to 5 = *very much*, similar to Holländer et al. (2019).

Explanation satisfaction. To validate how well the explanations described the automation's behavior and thus how satisfied participants were with the explanations, we used the Explanation Satisfaction Scale developed by Hoffman et al. (2023). The scale consists of seven items, such as "From the explanation I know how the software works", which participants rated on a 5-point Likert-scale from *I agree strongly* to *I disagree strongly*.



Fig. 1. Overview of the four traffic situations that participants experienced in each of the rides during the experiment.

Perceived safety. Currently, no questionnaire is applicable to measure *Perceived Safety* in the automotive context. While there are a few tools that measure this construct for different application domains or within social contexts (Schuss et al., 2023; Tzouras et al., 2020), a fully developed and validated scale for interactions with automation in the context of AD is still missing. Thus, we adapted items that were suggested by Nordhoff et al. as well suited to measure perceived safety (Nordhoff et al., 2021). The three items “*I felt relaxed during the ride*”, “*I felt safe during the ride*”, and “*I felt anxious during the ride*” were presented on a 5-point Likert scale ranging from 1 (not at all) to 5 (strongly).

Desire for control and perceived level of being in control. In regard to the psychological concept of *Sense of Control* as explained in Section 2, we measured participants’ desire for control in the vehicle, as well as their subjective perceived level of being in control. Measuring the desire to control can indicate the degree to which passengers agree with the automation’s behavior, as well as the explanations it provides. Additionally, assessing the control perception allows us to better understand the connection between AI explanations and the *Illusory Control bias* (Yarritu et al., 2014).

To measure participants’ desire for control, we used the single-item question “*I want to have control over the vehicle*”, which participants rated on a 5-point Likert scale from “*not at all*” to “*very much*”. Similarly, participants’ *feeling of being in control* was rated on the single-item question “*I feel like I have control in this situation*”, on a 5-point Likert scale from “*not at all*” to “*very much*”.

Post-experiment interview. After the participants had completed all experimental conditions within the VR simulation, we conducted a semi-structured interview to elicit their subjective perceptions of the different AI explanations. The participants were asked to rank the explanations according to their preference and to suggest any ideas on how the explanations could be improved. Concerning the sense of control, they were asked if there were situations where they wanted to intervene in the driving task and if they thought presenting the explanations influenced their perception of being in control over the situation.

4.4. Apparatus

The driving simulation experiment was conducted in VR. The experimental setup involved a gaming car seat in an empty laboratory. The urban driving environment was created and modeled using *Unity*

Technologies (2022). For the vehicle model, we chose a Volvo XC60. To simulate the physical movements of the Level 4 AV within the VR environment, we utilized IPG Carmaker (IPG Automotive GmbH, 2023), which was connected to Unity via UDP. The HTC Vive Pro Eye (HTC Corporation, 2025) was used as the VR headset, with the VR scene streamed from Unity through StreamVR (Valve Corporation, 2023). Fig. 3 illustrates the VR environment from the driver’s perspective.

The explanations were displayed on a black, translucent box in front of the driver (Fig. 2). The explanation display was activated during traffic situations in which the automation communicated the reasoning behind its decisions and was turned off during the ride between scenarios.

4.5. Procedure

After the welcome, participants were asked to fill out an informed consent form that detailed all the information and risks associated with participation. After they signed the form, they were provided with detailed information about the study procedure, and they filled out the demographic questionnaire. Participants then put on the VR glasses and were allowed to familiarize themselves with the HMD and the virtual environment. This was followed by the experiment runs. Participants experienced four rides in permuted order. In each of the rides, they experienced one of the three explanation conditions (“*How*”, “*How & Why*”, “*Contrastive & Counterfactual*”), and a “*Baseline*” condition with no explanations. Each ride lasted approximately five minutes and contained all four traffic scenarios described in Section 4.2 in randomized order. During the ride, after each traffic scenario, we verbally asked participants to rate their perceived level of being in control and their desire for control on a 5-point Likert Scale. After the ride, the described questionnaires or items for *Acceptance*, *User Experience*, *Trust*, *Explanation Satisfaction*, and *Perceived Safety* were filled out on a tablet screen. The next ride continued with the same procedure until all four rides with the four different explanation levels were presented in a randomized order. Following this part, the study concluded with an interview in which participants could more openly express their attitudes toward the different explanation levels. Instead, the researcher directly transcribed the participants’ feedback. For an engaging and deep discussion during the final interview, we showed printed versions of the explanation texts. Overall, the study took approximately 60 min per person, and participation was compensated with 15€. The research was conducted in accordance with the ethical standards outlined in

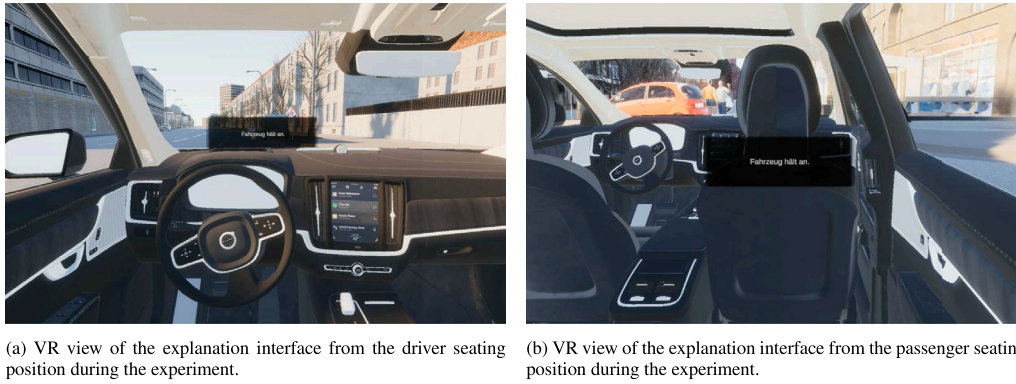


Fig. 2. Participant's perspective of the driving simulation and the explanation interface during the experiment. Depicted are the construction site situation (left) and the pedestrian on the street situation (right).



Fig. 3. Participant experiencing one of the scenarios inside the VR simulation.

the Declaration of Helsinki (World Medical Association, 2009), and ethics board approval was granted under the sign GEHBa-202311-V-140. Written informed consent was taken from participants for their anonymized data to be stored and analyzed. No physical or psychological harm was inflicted on the participants during the study. Participants were free to withdraw from the study at any time without a reason.

4.6. Participants

We recruited a total of $N = 50$ participants via internal and external mailing lists. The study included 25 drivers and 25 passengers (between-subject variable). All participants resided in Germany. Their ages ranged from 18 to 74 years, with an average age of 32.98 years ($SD = 16.35$). Of the participants, 31 were male, 17 were female, and two opted not to disclose their gender. All participants held a valid driver's license and reported driving an average of 8681.91 km per year.

5. Results

Quantitative data analysis was conducted using R (R Core Team, 2021). A significance level of $p = .05$ was used for all statistical analyses. For non-parametric factorial data, the Aligned-Rank-Transform (ART) Procedure (Wobbrock et al., 2011) was employed, and the Wilcoxon test was used to analyze ordinal questionnaire data. P-values were adjusted for multiple testing using the Bonferroni-Holm

method (Holm, 1979). Data from five participants (test subject numbers 19, 20, 30, 47, and 50) were excluded from the analysis due to their inability to fully understand the assigned task, resulting in a final sample size of $N = 44$.

5.1. Quantitative results

5.1.1. Desire for control

The ART analysis for participants' **Desire for Control** showed a significant main effect of the explanation level $F(3,129) = 3.99$, $p = .01$, while the main effect of the seating position was not significant $F(1,43) = 2.63$, $p = .112$. Similarly, there was no significant interaction effect of the explanation level and the seating position ($F(3,129) = .52$, $p = .672$). Post hoc tests using Wilcoxon Signed-Rank tests did not reveal significant differences between individual groups. More detailed results are presented in Fig. 4.

5.1.2. Feeling of being in control

The ART analysis for the **Perceived Level of Being in Control** showed a significant main effect of the explanation level $F(3,129) = 5.40$, $p = .002$, while the main effect of the seating position was not significant $F(1,43) = 1.02$, $p = .317$. Similarly, the analysis of the interaction effect of explanation condition and seating position did not show a significant effect $F(3,129) = 1.30$, $p = .279$. However, post hoc analysis using the Wilcoxon Signed-Rank test did not show any

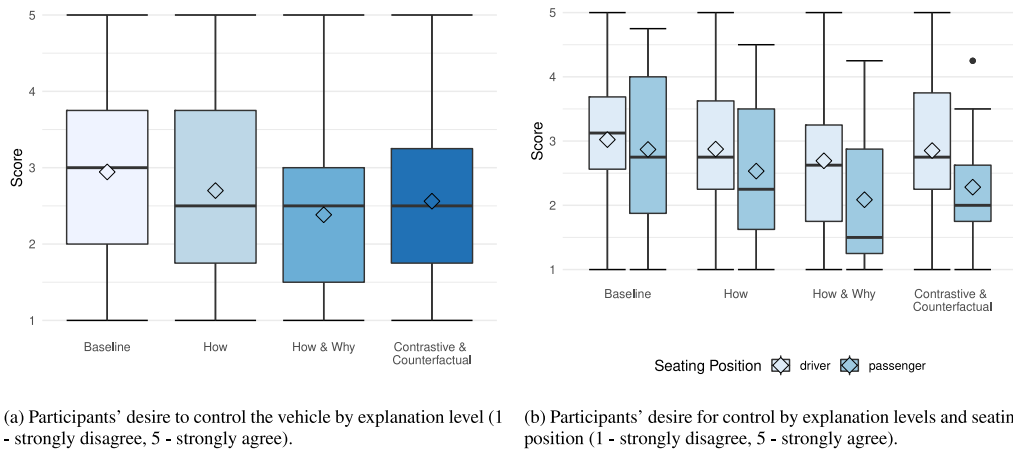


Fig. 4. Participants' desire to control the vehicle. The item “*I want to control the vehicle in this situation*” was rated on a 5-point-Likert-scale from “strongly agree” to “strongly disagree”.

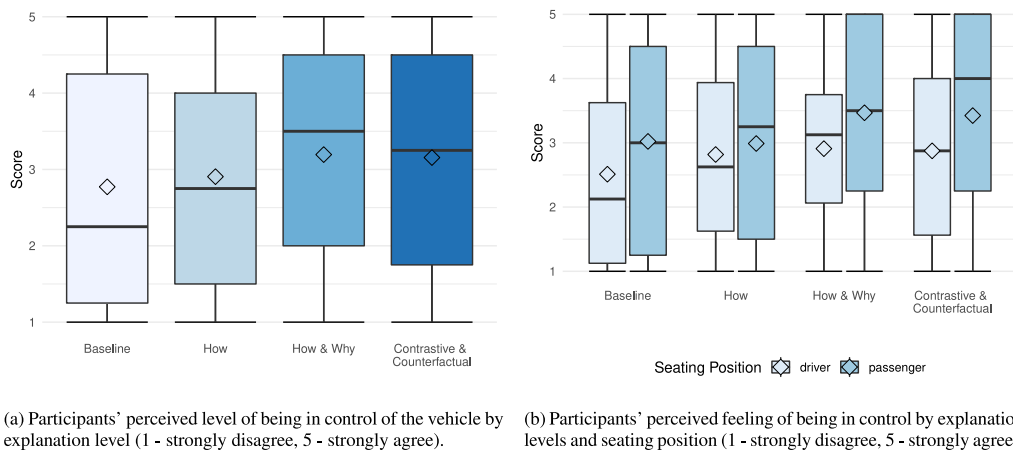


Fig. 5. Participants' perceived feeling of being in control. The item “*I feel like I am in control of the situation*”. was rated on a 5-point-Likert-scale from 1 - “strongly disagree” to 5 - “strongly agree”.

significant differences between individual groups. Detailed results are presented in Fig. 5.

5.1.3. User experience

Evaluating **Overall User Experience** using the ART procedure resulted in a significant main effect of the explanation level $F(3, 108) = 6.87, p < .001$ (Fig. 6). We did not find a significant main effect of seating position $F(1, 36) = .71, p = .405$ (Fig. 7) and similarly, the interaction effect of explanation level and the seating position was not significant $F(3, 108) = 2.63, p = .054$. Post hoc analysis for **Overall User Experience** using Wilcoxon Signed Rank tests found significant differences between the “Baseline” and “How & Why”, as well as the “Contrastive & Counterfactual” conditions. Also, we found a significant difference between the “How” and “How & Why” conditions. The analysis for **Pragmatic Quality** showed similar results, with a significant main effect of explanation level $F(3, 108.47) = 20.42, p < .001$, but no significant main effect of the seating position $F(1, 36), p = .018$, and a not significant interaction effect $F(3, 108.50), p = .842$. Post hoc analysis of the **Pragmatic Quality** revealed significant differences between the “Baseline” and “How”, “How & Why”, and “Contrastive &

Counterfactual” conditions. Similarly, we found significant differences between the “How” and “How & Why” as well as the “How & Why” and “Contrastive & Counterfactual” conditions. The same is true for **Hedonic Quality**, with a significant main effect of the explanation level $F(3, 108.24) = 6.87, p < .001$, but no significant main effect of the seating position $F(1, 36) = .71, p = .401$, or interaction effect $F(3, 108.24) = 2.63, p = .054$. Post hoc analysis only found a significant difference between the “Baseline” and “Contrastive & Counterfactual” conditions. Descriptive results for **User Experience** are presented in Figs. 6 and 7. A full list of the descriptive analysis of the UEQ-S, as well as the results for the post-hoc analysis, including significances, is listed in Table B.3 in Appendix B.1.

5.1.4. User acceptance

ART analysis for the acceptance scale revealed a significant main effect of the explanation condition on **Overall Acceptance** $F(3, 29.61) = 141.29, p < .001$. While the main effect of the seating position was not significant $F(1, 1.18) = 47.00, p = .282$, as the interaction effect between the explanation condition and the seating position $F(3, 1.37) = 141.32$,

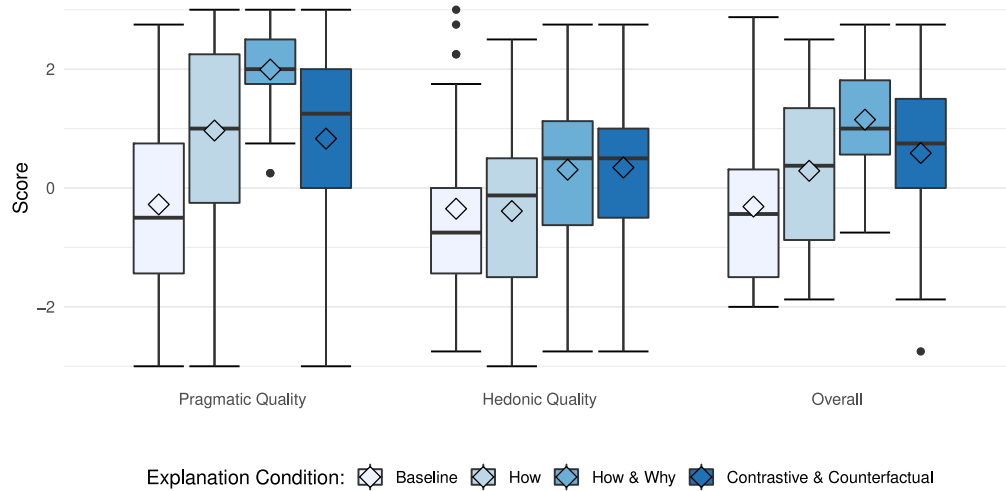


Fig. 6. Results for the User Experience Questionnaire short version for explanation conditions and the subscales *Pragmatic Quality* and *Hedonic Quality*.



Fig. 7. Results for the User Experience Questionnaire short version for participants' seating position and the subscales *Pragmatic Quality* and *Hedonic Quality*.

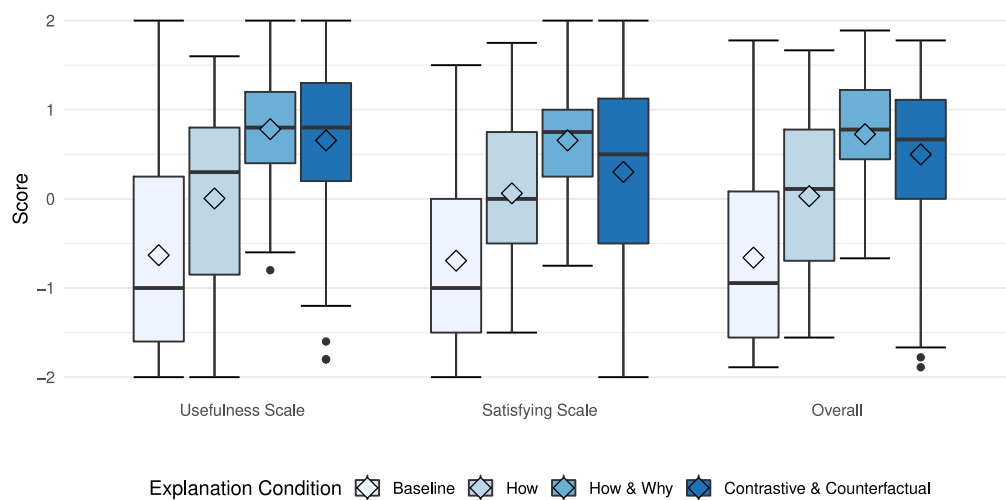


Fig. 8. Results for the acceptance scale by van der Laan et al. (1997), with the subscales *Usefulness* and *Satisfaction*.

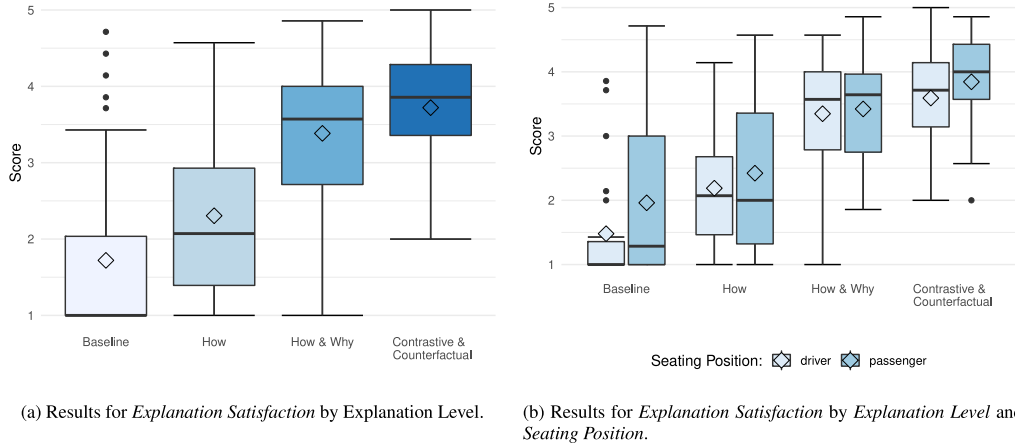


Fig. 9. Participants' subjective *Explanation Satisfaction* rated on the Explanation Satisfaction Scale by Hoffman et al. (2023), on a 5-point-Likert scale from 1 - "strongly disagree" to 5 - "strongly agree".

$p = .255$. Post hoc comparisons for the factor explanation levels revealed significant differences between the "Baseline" condition and the "How" condition, as well as the "How & Why" condition and the "Contrastive & Counterfactual" condition. There was also a significant difference between the "How" and "How & Why" conditions. We did not find a significant difference between any other groups. Similarly to the **Overall Acceptance**, the ART analysis for the **Usefulness** subscale showed a significant effect of the main effect of the explanation condition $F(3,30.61) = 141.27$, $p < .001$, but neither the seating position $F(1,34) = 47.00$, $p = .55$, or the interaction effect of explanation condition and seating position $F(3,1.50) = 141.30$, $p = .217$. Post hoc analysis using Wilcoxon Signed Rank tests revealed significant differences between the "Baseline" condition and the "How", "How & Why" condition, and the "Contrastive & Counterfactual" condition. There were also significant differences between the "How" condition and the "How & Why" condition. Lastly, analyzing the results of the **Satisfying** subscale, again we found a significant main effect of the explanation condition $F(3,24.33) = 141.31$, $p < .001$, but no significant main effect of the seating position $F(1,2.19) = 47.00$, $p = .145$, and no significant interaction effect $F(3,1.00) = 141.32$, $p = .393$. Results of the post hoc analysis using Wilcoxon Signed Rank tests revealed significant differences between the "Baseline" condition and the "How" condition, as well as the "How & Why" condition, and the "Contrastive & Counterfactual" condition. Also, we found a significant difference between the "How" and "How & Why" conditions. Descriptive results for *User Acceptance* are presented in Fig. 8. A full list of the descriptive analysis of user acceptance, as well as the results for the post hoc analysis, including significances, is listed in Table B.4 in Appendix B.2.

5.1.5. Explanation satisfaction

By calculating the Cronbach's alpha for the 9-item Explanation Satisfaction Scale by Hoffman et al. (2023), which we used in this experiment, we found excellent reliability of $\alpha = .95$. Thus, we assume that the scale is appropriate for measuring participants' satisfaction with the explanations.

ART analysis of participants' Explanation Satisfaction showed a significant main effect of the explanation level $F(3,126.27) = 58.17$, $p < .001$, while the main effect of seating position was not significant $F(1,42) = 1.28$, $p = .265$. Similarly, there was no significant interaction effect $F(3,126.27) = .34$, $p = .794$. Post hoc analysis using Wilcoxon Signed Rank tests showed significant differences between the "Baseline" ($Mdn = 1$, $IQR = 1.04$) and "How" condition ($Mdn = 2.07$, $IQR = 1.54$, $p = .002$, $r = .39$), "How & Why" condition ($Mdn = 3.57$,

$IQR = 1.29$, $p < .001$, $r = .61$), and "Contrastive & Counterfactual" condition ($Mdn = 3.72$, $IQR = .93$, $p < .001$, $r = .69$). Similarly, it found significant differences between the "How" and "How & Why" ($p < .001$, $r = .47$), as well as the "Contrastive & Counterfactual" condition ($p < .001$, $r = .59$). There was no significant difference between the "How & Why" and "Contrastive & Counterfactual" conditions. The results are shown in Fig. 9

5.1.6. Perceived safety & trust in automation

An ART analysis showed a significant main effect of the explanation level on participants' **Perceived Safety** $F(3,126.09) = 13.79$, $p < .001$, while the main effect of seating position was not significant $F(1,42) = .04$, $p = .844$, as well as the interaction effect $F(3,126.10) = 1.49$, $p = .221$ (Fig. 10 (a)). Post hoc analysis using Wilcoxon Signed Rank tests, only revealed significant differences between the "Baseline" ($Mdn = 2.83$, $IQR = 2$) and "How & Why" ($Mdn = 4.00$, $IQR = 1.67$, $p = .015$, $r = .32$), as well as the "Contrastive & Counterfactual" conditions ($Mdn = 3.67$, $IQR = 1.5$, $p = .023$, $r = .31$).

Analyzing **Trust in Automation** revealed a significant main effect of the explanation level $F(3,125.16) = 32.13$, $p < .001$, while the main effect of the seating position was not significant $F(1,41.99) = 1.95$, $p = .169$. The interaction effect between the explanation level and the seating position also was not significant $F(3,125.18) = 2.57$, $p = .058$ (Fig. 10(b)). Post hoc analysis revealed significant differences between the "Baseline" ($Mdn = 2.00$, $IQR = 2.25$) and "How & Why" ($Mdn = 4.00$, $IQR = 1$, $p < .001$, $r = .57$), and "Contrastive & Counterfactual" ($Mdn = 4.00$, $IQR = 1$, $p < .001$, $r = .55$) conditions. Similarly we found a significant difference between the "How" and "How & Why" ($p < .001$, $r = .43$), as well as the "Contrastive & Counterfactual" ($p < .001$, $r = .42$) conditions.

5.2. Qualitative results

5.2.1. Subjective preference

As part of the post-experiment interviews, we asked participants to rank the three explanation levels based on their personal preference, with their favorite explanation concept in first place and their least favorite in the last place. Analyzing these rankings using the Borda-Count method, points are attributed according to the ranking of the concept in reverse order. The results are shown in Table 1. Thus, the "How & Why" explanation concept was preferred by participants the most overall, while the other two concepts are tied to personal preference.

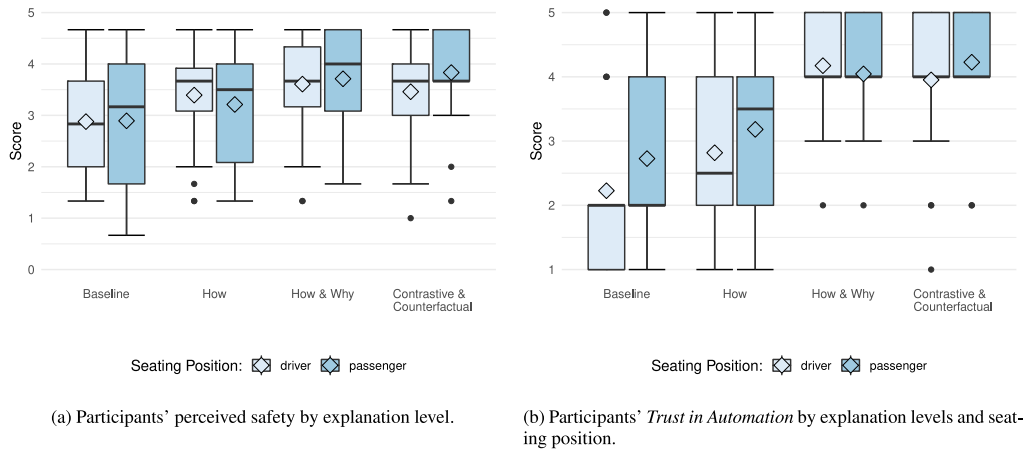


Fig. 10. Results for participants' *Perceived Safety* and *Trust in Automation*.

Table 1

Results for the Borda-Count analysis for personal preference of the explanation concepts.

"How & Why"	129 points
"Contrastive & Counterfactual"	65 points
"How"	65 points

5.2.2. Information content

Analyzing the semi-structured post-experiment interview, 48 participants mentioned that they generally wish for some kind of explanation from the automation during an automated ride. However, the explanation level had a significant impact on how useful or appropriate participants perceived the explanation. Thus, 38 participants mentioned that the "How" explanations were not helpful, with 18 participants noting that these explanations only state what they are easily able to perceive themselves without any additional explanation by the automation. Thus, 11 participants found it more hindering or annoying rather than helpful. Only a small minority found the information content sufficient ($n = 8$).

The "How & Why" concept received more positive feedback, with 39 participants stating that the information content was sufficient, and $n = 32$ participants even mentioned it to be their favorite explanation concept. However, six participants, on the other hand, were still dissatisfied with the explanation content and wished for more detailed information about the vehicle's surroundings (detected objects, etc. $n = 6$), timing information (when is the vehicle able to continue the ride? $n = 2$), etc., or did not understand the explanations ($n = 2$).

The information content of the "Contrastive & Counterfactual" explanations was perceived as too detailed by 36 participants. Primarily, participants were dissatisfied with the written presentation of the information. They would prefer a visual representation of the information in the form of icons or symbols ($n = 16$), auditory signals through a voice interface ($n = 9$), or simply better descriptions in the explanations overall ($n = 14$). As another aspect, five participants noted that they would like to be able to customize the explanations in terms of personalized information content or have the option to deactivate the explanation GUI completely.

5.2.3. Sense of control: Desire for control and perceived level of being in control

Participants mentioned that they felt the different explanation conditions had an impact on their perceived level of control during the ride. Even though they did not have the option to intervene or control

the vehicle during the experiment, $n = 25$ participants mentioned that the increased level of detail in the explanation in the "How & Why" and "Contrastive & Counterfactual" conditions did increase their subjective feeling of having control during the ride. Since they were better able to understand the automation's decisions, they, in return, felt more in control. However, a large portion of participants stated that they had not noticed such an effect ($n = 16$).

However, 42 participants reported feeling the urge to intervene in the driving task on some occasions during the ride. Mostly because they disagreed with the automation's behavior and were still dissatisfied with the explanation the automation provided as a rationalization. For a method of intervention, 22 participants would prefer to take full control over the vehicle using conventional controls (steering wheel and pedals). In comparison, nine participants would only need the pedals to brake or accelerate. Alternatively, 15 participants preferred a command-based interface, in which they could choose to execute specific commands by pressing a button.

6. Discussion

This research contributes significant empirical evidence to the growing field of Explainable AI (XAI) in AD, particularly in terms of how the level of explanation detail influences user trust, perceived control, and system acceptance. In our experiment, we found that the explanation level influenced every measurement assessed in the study. Overall, the results show that more information improved *User Experience*, *Acceptance*, *Trust*, and *Perceived Safety*. However, it is also evident that the "Contrastive & Counterfactual" condition did not yield improved results over the "How" or "How & Why" conditions. Instead, participants perceived the information content of this condition as being too detailed in many instances. Thus, the trend of improving values with higher explanation content turns back down towards the "Contrastive & Counterfactual" condition.

6.1. The impact of explanation level on user perception

Echoing previous work on XAI in AD (Koo et al., 2014; Zhang et al., 2021), our results confirm that explanations improve user experience and trust, but also underline the importance of a calibrated level of detail. The medium level "How & Why" explanations struck the most effective balance, concise yet informative. These explanations were rated significantly higher than both the minimalist "How" condition and the over-detailed "Contrastive & Counterfactual" condition, confirming prior concerns about cognitive overload with verbose or

complex explanations (Koo et al., 2014; Omeiza et al., 2021; Miller, 2019). However, the preferred level of detail seems to be influenced by the complexity or dynamics of the information. Colley et al. (2020) compared explanation levels (named clarity states) in the context of pedestrian intention and found that the highest clarity state was the most preferred option, leading to lowest cognitive load ratings, highest trust and highest capability ratings.

While Stepin et al. (2021) highlight the potential of counterfactual reasoning for transparency and user empowerment, our findings suggest that, in real-time AV contexts, this type of explanation may not translate well into practice when presented in verbose textual form. Participants in our study frequently cited the excessive length and lack of intuitiveness of these high-detail explanations. This is consistent with previous research in AD that emphasizes the need for modality-sensitive and time-aware explanation delivery in dynamic settings (Schneider et al., 2021; Kim et al., 2023).

The “How” explanations, while still outperforming the baseline, were often perceived as redundant, a sentiment supported by Koo et al. (2014), who also noted that purely descriptive statements fail to enhance users’ mental models or understanding. Participants in our study indicated that explanations need to move beyond what is already observable to be seen as valuable.

6.2. Illusion of control and perceived control

A notable contribution of this study is the empirical validation of the Illusion of Control bias in the context of AVs. Our data show that as explanations became more detailed, participants’ perceived control increased while their desire for control decreased, despite having no actual influence over the vehicle. This aligns with findings by Koo et al. (2016) and Manger et al. (2023b), and further strengthens the argument that explanations moderate not just trust, but also the user’s subjective sense of agency.

Interestingly, while increased perceived control may enhance comfort and reduce anxiety, it also poses potential risks. If users overestimate their influence or understanding of the system due to well-crafted explanations, this may lead to overtrust or complacency, a phenomenon previously described by Lee and See (2004a) and Muir and Moray (1996). Thus, explainable AI in AVs must balance transparency and bounded realism, providing insight without inducing false beliefs about control or system capabilities.

6.3. Seating position and role perception

Contrary to our expectations and partially addressing RQ3 and RQ4, the seating position did not have a statistically significant effect on participants’ perception of control or user experience measures. However, qualitative data suggest that seating may still subtly prime users’ expectations. Participants in the driver’s seat reported a slightly greater desire for control, potentially due to ingrained role expectations associated with that position (Malve et al., 2023a; Peintner et al., 2023b; Guo et al., 2019).

This subtle priming effect mirrors past findings that suggest the context of control, beyond actual control, can shape cognitive and emotional responses (Wolf, 2016; Malve et al., 2023a). Although the quantitative results were not conclusive, future research should investigate how the physical environment and role framing shape expectations of control, especially in shared mobility contexts.

Although evidence on seating position differences in information needs is limited, studies suggest that the type and amount of information can help maintain trust and support engagement in non-driving activities. Prior research shows that augmenting otherwise invisible traffic participants (e.g., in dense fog) or presenting upcoming maneuvers while seated backwards can increase acceptance and trust (Wintersberger et al., 2019). Therefore, we emphasize the importance of providing rear-seated passengers with sufficient information about the vehicle’s behavior.

6.4. Toward more effective explanation modalities

One of the most prominent takeaways from the qualitative responses is that text alone is insufficient in dynamic driving environments. Participants consistently expressed a preference for multimodal explanations, including icons, auditory cues, and interactive controls. This resonates with recent research emphasizing the importance of multimodal and adaptive HMIs in AVs (Peintner et al., 2023b; Wintersberger et al., 2020).

Moreover, the “one-size-fits-all” approach to explanation granularity is insufficient. Preferences varied widely across participants, with some requesting more detail while others found it overwhelming. This supports the growing body of work advocating for personalized and context-aware explainability (Wiegand et al., 2020b; Danner et al., 2021), where users can dynamically adjust the depth and format of explanations according to their needs.

Peng et al. (2024) showed that to enhance comfort in AVs, designers should consider suitable communication channels, as well as ensure that the AV’s capabilities match the user’s expectations. Lack of control over the vehicle was categorized as a source of discomfort, as well as unfulfilled expectations, which included unexpected operations, such as driving too slowly. In contrast to “communication” as a source of comfort, “poor communication” emerged as a source of discomfort, capturing users’ lack of understanding regarding the AV’s capabilities and maneuvers.

7. Limitations

In the presented study, one major limitation, which also became apparent in the results, was the use of only text-based explanations. In the context of the study, this decision was made to ensure comparability between the different explanation strategies. However, in real-world applications, using various modalities for communicating AI explanations is more suitable, especially in very dynamic contexts, such as automated mobility. Reading lengthy texts in critical driving situations is far from ideal and was mentioned by participants as a hindrance, which contributed to the lower ratings of the counterfactual explanations. Thus, future research should focus on identifying the most efficient and effective communication strategies for conveying AI explanations.

Furthermore, since AVs at level 4 are not currently available in Germany, we utilized a VR driving simulator for our experiment. While immersion using VR can be seen as sufficient for this application, differences in real-world situations can be expected, as higher driving simulator fidelity can often be linked to more valid results; however, behavioral differences are not as pronounced (Himmels et al., 2024).

Regarding the “Level of Explainability”, more diverse traffic situations should be investigated in future research. Although in this study, the highest “Level of Explainability”, “Contrastive & Counterfactual” explanations, were perceived as too much information, there might be situations, such as safety-critical incidents, in which the AV needs to communicate more detailed and complex information.

8. Implications for future research and design

This work directly supports and extends the growing subfield of explainability in automotive HMI design by providing empirical thresholds for explanation effectiveness. The identification of “How & Why” explanations as a sweet spot for user satisfaction and trust without overloading cognitive resources contributes to a more refined understanding of user-centered design principles for AD systems.

Overall, our findings suggest that higher explainability is not necessarily beneficial. In real-world driving contexts, explanations must be informative, relevant, and easily digestible in real-time. Designers should prioritize mid-level explanations “How & Why”, which users perceive as sufficiently transparent without being cognitively taxing.

Additionally, the study strengthens the argument for multimodal, adaptive explanations, aligning with calls for context-sensitive and personalized interfaces in AVs. It also challenges the field to consider non-visual explanation delivery, particularly when users are distracted or engaged in NDRTs, a realistic use case for fully AVs.

Therefore, future research should explore alternative modalities for delivering high-detail explanations, primarily visual and auditory formats that minimize cognitive load. Moreover, it is crucial to incorporate NDRTs into experimental designs, as these represent realistic distractions that affect both attention and trust calibration. Given that many participants stated their desire for control would decrease during NDRTs, understanding how explainability operates in multitasking contexts is essential.

Finally, the observed *Illusion of Control* raises ethical questions for researchers and developers: if explanations can artificially inflate perceived control, how should we regulate and calibrate such perceptions to ensure user safety without undermining trust?

For industry stakeholders, including automotive manufacturers, HMI designers, and AI system developers, our results offer practical guidance for the deployment of AV explanation systems:

- **Avoid minimalist or overly verbose explanations:** Users distrust simple, observable feedback (“How” explanations) and feel overwhelmed by dense, counterfactual explanations. Mid-level, “How & Why” explanations balance informativeness with digestibility.
- **Support perceived safety and trust through targeted transparency:** Well-designed explanations can reduce users’ desire for control, potentially mitigating unnecessary human intervention and improving ride comfort.
- **Rethink rear-seat UI design:** Visibility and interaction limitations in passenger seating must be addressed. Visual obstructions reduce the utility of explanations, highlighting the need for voice or augmented displays for passengers.
- **Design for realistic scenarios:** Since most AV use involves NDRTs (e.g., phone use, entertainment), industry solutions must convey critical information without requiring full visual attention.

9. Conclusion

This work investigated the relationship between the explainability of AVs and passengers’ *Sense of Control*, as well as their perception of the vehicle regarding *Users’ Acceptance*, *Perceived Safety*, *Trust in Automation*, and *User Experience*. We also examined *Explanation Satisfaction* to gauge how well they received the automation’s explanations. In a virtual reality study, $N = 44$ participants were exposed to various urban driving scenarios and presented with an HMI that provided three levels of explainability (low, medium, high) regarding the vehicle’s decisions in written form, along with a baseline condition where no information was provided. The findings suggest that higher levels of explainability generally enhance *User Experience*, *Trust*, and *Perceived Safety*. Still, there seems to be a threshold beyond which more detailed explanations can become overwhelming and counterproductive, as seen in the highest “*Level of Explainability*” by providing “*Contrastive & Counterfactual*” explanations. Overall, participants were more satisfied with explanations, which contained a greater level of detail, with the “*Contrastive & Counterfactual*” condition receiving the highest ratings. The “*Level of Explainability*” also had a significant impact on participants’ desire for control. Higher explanation levels generally led to an improved perception of control and, in turn, a reduced desire for control.

Interestingly, while explainability levels significantly influenced participants’ perceived level of being in control and satisfaction with the AV’s decisions, the seating position (driver vs. passenger) did

not have a substantial impact on these factors. This implies that the effectiveness of explanations is largely independent of the passengers’ ability to observe the traffic situation directly, although qualitative feedback indicated that visibility issues in the passenger position could diminish the perceived value of less detailed explanations.

Overall, this research highlights the importance of tailoring the level of detail in AI explanations to optimize user satisfaction and trust, without overwhelming passengers with excessive information. Future work could explore alternative methods of delivering complex explanations, such as through visual or auditory aids, to better support user comprehension and engagement in dynamic driving environments.

CRediT authorship contribution statement

Jakob Peintner: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Carina Manger:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Melanie Berger:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Andreas Riener:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Method

A.1. Traffic situations and explanation texts

See [Table A.2](#).

Appendix B. Post-hoc test results

B.1. User experience

See [Table B.3](#).

B.2. User acceptance

See [Table B.4](#).

Data availability

Data will be made available on request.

Table A.2

Overview of the AI explanations for each of the four traffic situations.

Traffic situation		Level of explainability			
Nr.	Name	(1) No explanation	(2) Low explanation - how	(3) Medium explanation - how & why	(4) High explanation - contrastive & counterfactual
1	Slow-moving vehicle ahead (SMV)	...	Vehicle slows down.	Slow vehicle ahead. Vehicle slows down.	Vehicle cannot drive faster because a slow vehicle ahead was detected. It will accelerate if the vehicle ahead drives faster or leaves the lane.
2	Pedestrians talking and standing on the road (POR)	...	Vehicle stopped.	Pedestrians detected. Vehicle stopped.	Vehicle cannot drive on because pedestrians on the street were detected. It will drive on if pedestrians leave the street.
3	Pedestrian with unclear intention standing next to the crosswalk (PUI)	...	Vehicle stopped.	Pedestrian detected. Vehicle stopped.	Vehicle cannot drive on because pedestrian with unclear intention was detected. It will drive on if the pedestrian does not cross the road.
4	Construction Site (COS)	...	Vehicle stopped.	Blocked lane ahead. Vehicle stopped.	Vehicle cannot drive on because the lane is blocked by a construction site. It will continue once the situation has cleared up.

Table B.3

Pairwise comparisons for user experience depending on the explanation condition using the Wilcoxon Signed Rank test.

Group 1	Mdn	IQR	Group 2	Mdn	IQR	W	p	p-adjusted	Sig.	Effect-size	
Overall User Experience											
Baseline	-.44	1.81	How	.38	2.22	515	.032	.19	ns	.25	small
Baseline	-.44	1.81	How & Why	1	1.25	255	< .001	< .001	****	.57	large
Baseline	-.44	1.81	Cont. & Countf.	.75	1.5	424	.003	.019	*	.34	moderate
How	.38	2.22	How & Why	1	1.25	456	.004	.023	*	.33	moderate
How	.38	2.22	Cont. & Countf.	1.25	2	607	.31	1	ns	.12	small
How & Why	1	1.25	Cont. & Countf.	1.25	2	859	.15	.92	ns	.16	small
Pragmatic Quality											
Baseline	-.5	2.19	How	1	2.5	404	< .001	.006	**	.38	moderate
Baseline	-.5	2.19	How & Why	2	.75	164	< .001	< .001	****	.67	large
Baseline	-.5	2.19	Cont. & Countf.	1.25	2	424	.003	.019	*	.34	moderate
How	1	2.5	How & Why	2	.75	436	.002	.011	*	.36	moderate
How	1	2.5	Cont. & Countf.	1.25	2	694	.932	1	ns	.01	small
How & Why	2	.75	Cont. & Countf.	1.25	2	1036	.001	.006	**	.38	moderate
Hedonic Quality											
Baseline	-.75	1.44	How	-.125	2	680	.665	1	ns	.05	small
Baseline	-.75	1.44	How & Why	.5	1.75	507	.017	.102	ns	.27	small
Baseline	-.75	1.44	Cont. & Countf.	.5	1.5	436	.005	.027	*	.33	moderate
How	-.125	2	How & Why	.5	1.75	546	.047	.283	ns	.23	small
How	-.125	2	Cont. & Countf.	.5	1.5	509	.04	.239	ns	.24	small
How & Why	.5	1.75	Cont. & Countf.	.5	1.5	729	.942	1	ns	.01	small

Table B.4

Pairwise comparisons for user acceptance depending on the explanation condition using the Wilcoxon Signed Rank test.

Group 1	Mdn	IQR	Group 2	Mdn	IQR	W	p	p-adjusted	Sig.	Effect-size	
Overall User Acceptance											
Baseline	-.94	1.64	How	.11	1.47	714	< .001	.003	**	.35	moderate
Baseline	-.94	1.64	How & Why	.78	.78	354	< .001	< .001	****	.61	large
Baseline	-.94	1.64	Cont. & Countf.	.67	1.11	531	< .001	< .001	****	.47	moderate
How	.11	1.47	How & Why	.78	.78	697	< .001	.001	**	.37	moderate
How	.11	1.47	Cont. & Countf.	.67	1.11	902	.289	1	ns	.2	small
How & Why	.78	.78	Cont. & Countf.	.67	1.11	1396	.99	.92	ns	.14	small
Usefulness Scale											
Baseline	-1	1.85	How	.3	1.65	776	.003	.015	*	.3	moderate
Baseline	-1	1.85	How & Why	.8	.8	370	< .001	< .001	****	.6	large
Baseline	-1	1.85	Cont. & Countf.	.8	1.1	494	< .001	< .001	****	.5	large
How	.3	1.65	How & Why	.8	.8	696	< .001	.001	**	.37	moderate
How	.3	1.65	Cont. & Countf.	.8	1.1	822	.011	.064	ns	.26	small
How & Why	.8	.8	Cont. & Countf.	.8	1.1	1322	.388	1	ns	.09	small
Satisfying Scale											
Baseline	-1	1.44	How	0	2	671	< .001	< .001	***	.38	moderate
Baseline	-1	1.44	How & Why	.75	1.75	392	< .001	< .001	****	.59	large
Baseline	-1	1.44	Cont. & Countf.	.5	1.5	634	< .001	< .001	***	.4	moderate
How	0	2	How & Why	.75	1.75	748	< .001	.005	**	.34	moderate
How	0	2	Cont. & Countf.	.5	1.5	1042	.237	1	ns	.11	small
How & Why	.75	1.75	Cont. & Countf.	.5	1.5	1449	.076	.453	ns	.18	small

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